

Simulation-Based Inference

Exercise 9

1. Train a neural network and compare design choices

Train a neural network on the Fashion-MNIST dataset which is a benchmark dataset for image classification, consisting of 70,000 grayscale images of ten categories of clothing items. The goal is to train a model that, given a new image of clothing, correctly predicts which of the ten categories it belongs to. Systematically investigate how the following design choices affect performance:

1. Define a simple 2-layer fully connected MLP with ReLU. Split the training set into train and val set with 50,000 training images and 10,000 validation images. Train for 20 epochs, plot train & val accuracy curves and calculate the test set accuracy for the following tasks:
2. Select best optimizer among SGD, RMSProp, Adam, by highest test accuracy.
3. Use the best optimizer from above and compare regularization techniques (no regularization, dropout, early stopping, and dropout + early stopping) by training for 50 epochs. Select the best regularization strategy by highest test accuracy.
4. Use the best optimizer and regularization method from above and compare learning-rate scheduling (step decay, cosine decay).
5. Vary model architecture (e.g. a simple MLP vs. a small convolutional network, different depths/widths). For each architecture, use best optimizer + best regularization + best LR schedule found so far. Train up to 50 epochs (with early stopping if enabled). Select the best architecture by highest test accuracy.

Note: You can use google colab which provides free GPU for a limited time for training. You can also use Kaggle which provides faster GPUs for 30 hours per week which is sufficient for this task.